

MediBridge – Software Engineering Use Cases

UC-01: User Registration and Login

Primary Actor: Patient / Doctor / Admin

Description:

Allows users to register for an account or log in to MediBridge using their credentials.

Preconditions:

- The user has access to the MediBridge website.
- For login, the user must already have an account.

Main Flow:

1. The user opens the MediBridge login or registration page.
2. The user enters the required information.
3. The system validates the information.
4. The system authenticates the user.
5. The system identifies the user's role.
6. The system redirects the user to the appropriate dashboard.

Postcondition:

- The user is successfully authenticated and can access role-specific features.

UC-02: Search and Filter Doctors

Primary Actor: Patient

Description:

Allows patients to search for doctors and filter the available doctors according to their requirements.

Preconditions:

- The patient is logged in.
- Doctor information is available in the system.

Main Flow:

1. The patient opens the doctor search section.
2. The patient enters a search term or selects filters.
3. The system searches the available doctor records.
4. The system displays matching doctors.

5. The patient selects a doctor to view more information.

Postcondition:

- The patient receives a list of suitable doctors.

UC-03: View Doctor Profile

Primary Actor: Patient

Description:

Allows a patient to view a doctor's professional information before booking an appointment.

Preconditions:

- The patient has searched for or selected a doctor.

Main Flow:

1. The patient selects a doctor.
2. The system retrieves the doctor's profile.
3. The system displays the doctor's relevant information.
4. The patient can proceed to book an appointment.

Postcondition:

- The patient has access to the selected doctor's profile.

UC-04: Book Appointment

Primary Actor: Patient

Description:

Allows patients to schedule an appointment with an available doctor.

Preconditions:

- The patient is logged in.
- The doctor exists in the system.
- Appointment slots are available.

Main Flow:

1. The patient searches for a doctor.
2. The patient views the doctor's profile.
3. The patient selects an available appointment slot.
4. The patient confirms the appointment.
5. The system records the appointment.

6. The system updates the appointment information.
7. The relevant user receives a notification.

Postcondition:

- The appointment is successfully created.

UC-05: Manage Appointments

Primary Actor: Patient / Doctor

Description:

Allows patients and doctors to view and manage their appointments.

Preconditions:

- The user is logged in.
- An appointment exists.

Main Flow:

1. The user opens the appointments section.
2. The system displays the user's appointments.
3. The patient can cancel or reschedule an appointment.
4. The doctor can confirm or cancel an appointment.
5. The system updates the appointment status.
6. The system reflects the updated appointment information.

Postcondition:

- The appointment has the latest status and schedule.

UC-06: Manage Medical Records

Primary Actor: Doctor

Secondary Actor: Patient

Description:

Allows doctors to create medical records and patients to view their medical information.

Preconditions:

- The doctor is authenticated.
- The patient exists in the system.

Main Flow:

1. The doctor selects a patient.

2. The doctor enters the required medical information.
3. The system validates the information.
4. The system saves the medical record.
5. The patient can subsequently access the record.

Postcondition:

- The medical record is stored and available to the appropriate patient.

UC-07: Manage Prescriptions

Primary Actor: Doctor

Secondary Actor: Patient

Description:

Allows doctors to add prescriptions and patients to view their prescriptions.

Preconditions:

- The doctor is logged in.
- The patient exists in the system.

Main Flow:

1. The doctor selects a patient.
2. The doctor enters prescription information.
3. The doctor submits the prescription.
4. The system validates and stores the prescription.
5. The prescription becomes available to the patient.

Postcondition:

- The prescription is successfully stored and can be viewed by the patient.

UC-08: Manage Medications and Track Intake

Primary Actor: Patient

Description:

Allows patients to manage their medications and track whether they have taken their medication.

Preconditions:

- The patient is logged in.

Main Flow:

1. The patient opens the medication section.
2. The patient adds or views medication information.
3. The patient records medication intake.
4. The system stores the medication activity.
5. The patient's medication tracking information is updated.

Postcondition:

- The patient's medication information and intake history are updated.

UC-09: View Care Score and Health Timeline

Primary Actor: Patient

Description:

Allows patients to view their healthcare progress through the Care Score and Health Timeline.

Preconditions:

- The patient is logged in.
- Relevant patient activity/data exists.

Main Flow:

1. The patient opens the Care Score or Health Timeline section.
2. The system retrieves the patient's relevant information.
3. The system calculates/displays the Care Score.
4. The system displays the patient's health timeline.

Postcondition:

- The patient can view their healthcare progress and timeline.

UC-10: Submit Doctor Review

Primary Actor: Patient

Description:

Allows patients to submit feedback or ratings for doctors.

Preconditions:

- The patient is logged in.
- A doctor exists in the system.

Main Flow:

1. The patient selects the relevant doctor.
2. The patient provides a rating or review.
3. The system validates the submitted information.
4. The system stores the review.
5. The review becomes associated with the doctor.

Postcondition:

- The patient's review is successfully recorded.

UC-11: Manage Doctor Availability

Primary Actor: Doctor

Description:

Allows doctors to manage the appointment slots in which they are available.

Preconditions:

- The doctor is logged in.

Main Flow:

1. The doctor opens the availability section.
2. The doctor selects available dates and time slots.
3. The doctor saves the availability.
4. The system updates the doctor's available slots.
5. Patients can see the available slots while booking appointments.

Postcondition:

- The doctor's availability is updated.

UC-12: Admin Management

Primary Actor: Admin

Description:

Allows the administrator to manage users, doctors, appointments and other system resources.

Preconditions:

- The administrator is authenticated.

Main Flow:

1. The admin logs into the admin dashboard.
2. The system displays system statistics and management options.

3. The admin manages patient accounts.
4. The admin manages doctor accounts.
5. The admin verifies doctors.
6. The admin manages appointments and specializations.
7. The admin manages healthcare resources.
8. The admin reviews reported issues.
9. The admin can view audit logs.

Postcondition:

- The required administrative information or system records are updated.

Summary of Use Cases

ID	Use Case	Primary Actor
UC-01	User Registration and Login	Patient / Doctor / Admin
UC-02	Search and Filter Doctors	Patient
UC-03	View Doctor Profile	Patient
UC-04	Book Appointment	Patient
UC-05	Manage Appointments	Patient / Doctor
UC-06	Manage Medical Records	Doctor / Patient
UC-07	Manage Prescriptions	Doctor / Patient
UC-08	Manage Medications and Track Intake	Patient
UC-09	View Care Score and Health Timeline	Patient
UC-10	Submit Doctor Review	Patient
UC-11	Manage Doctor Availability	Doctor
UC-12	Admin Management	Admin

Overall Use-Case Objective

The overall objective of these use cases is to define how the three major actors—**Patient, Doctor and Admin**—interact with the MediBridge system. The use cases describe the functional behavior of the system from the user's perspective and provide a foundation for system design, development and testing.

